Class Projects

Single-Molecule & Spatial Sequencing: Methods & Applications

I1: Building on the ideas for HW2, can you provide convincing evidence that the HCV virus contains modified nucleotides (either m6A, pseU, m5C, or inosine). This project involves implementing site-level filtering on the base modification calls, investigating possible site-level signal level visualization tools (such as methylartist: https://github.com/adamewing/methylartist), and implementing one of these to visualize putative modification sites [and a few non-modified sites].

I2: Examine annotated full length reads, along with truncated reads of similar kmers, to recognize current changes associated with 7-methyl guanosine. Can 7-methyl guanosine be detected in the Nanopore? Propose a strategy for developing a 7-methyl guanosine basecaller and possible significance.

I3: Building on recent advances in foundation models for biomedical imaging and omics, this project challenges you to explore the feasibility of training a multi-modal foundation model specifically designed for spatial omics images (e.g., CosMx, Xenium, MERSCOPE). While several recent methods (e.g., self-supervised learning on histopathology or transcriptomic image patches) have been proposed, current approaches are limited in generalizability, scalability, or modality integration.

I4: Using astro-STAMP PBMC datasets generated with the CosMx WTx assay, develop and implement a trajectory analysis framework to map dynamic changes in immune cells during and after spaceflight. Conventional single-cell trajectory methods (e.g., Monocle, Slingshot) assume dense sampling along a continuous differentiation axis and are not directly applicable to this dataset, which is sparse, discrete across timepoints, and spatially structured. Potentially, one could design a custom trajectory or temporal modeling strategy that can accommodate the specific properties of the astro-STAMP data. This can also be extended to multiple missions.

I5: Using astronaut datasets from astro-STAMP PBMC CosMx WTx (non-tissue, single-cell spatial transcriptomics), GeoMx multi-cell profiling (intact tissue), and optionally tissue-based CosMx WTx data from non-matched clinical samples, systematically compare the strengths and limitations of suspension and intact tissue spatial profiling technologies.

I6: Using SAHA data and any additional available datasets with both RNA and protein modalities (e.g., CosMx RNA+Protein, 10X Xenium with antibody panels), perform a systematic integration of transcriptomic and proteomic signals. Potential directions include benchmarking integration approaches (e.g., Seurat v5 multimodal integration, totalVI, MOFA+), identifying concordant and discordant signals between RNA and protein levels, and exploring biological insights uniquely revealed by integrative analysis (e.g., post-transcriptional regulation, differential protein expression not captured at RNA level).

I7: Building on datasets available for CosMx (1k panel, 6k panel, WTx panel) and Xenium (350-plex and 5k panel), perform a systematic evaluation of how panel size and platform choice affect biological discovery and data quality. This project involves subsetting datasets to matched panels, comparing sensitivity, specificity, and pathway coverage, and assessing how panel size impacts key analyses such as clustering, differential expression, and cell type annotation. Comparisons between CosMx and Xenium (5k vs 6k) should also be critically evaluated. Develop summary figures (e.g., gene detection vs panel size plots, heatmaps of pathway coverage) and propose panel/platform selection recommendations based on project goals.

I8: Investigate strategies for integrating multiple imaging-based spatial transcriptomics datasets into a 3D or multi-sample framework, especially across different platforms and resolutions (e.g., CosMx, Xenium, H&E, Antibody-based methods). This project involves identifying computational strategies for co-registration and integration (e.g., anchor-based alignment, mutual nearest neighbors, non-rigid registration), evaluating challenges such as platform-specific biases (e.g., signal intensity, resolution), and proposing workflows for generating unified 3D tissue reconstructions or multi-sample atlases.

I9: 5mC is a key epigenetic marker, repressing transcription and maintaining heterochromatin. By contrast, 5hmC is an epigenetic marker of active transcription. This project involves analyzing different 5hmC and 5mC from murine BMDM samples with either PBS-treatment, extracellular vesicle-DNA treatment, and lentiviral treatment. The goal is to perform different methylation analysis, in lieu of a different RNA-seq analysis.

I10: Building on the ideas for HW2, can you provide convincing evidence that the ZIKV virus contains modified nucleotides (either m6A, pseU, m5C, or inosine). This project involves implementing site-level filtering on the base modification calls and examining both putative modification and non-modified sites. The project also involves designing a proper k-mer specific negative control with non-virus matched IVT data.

I11: Nanopore direct-RNA sequencing captures single-molecule polyA tail length and epitranscriptome modifications. This project involves performing a gene-level single-molecule correlation analysis between polyA length and the presence of a particular modification on the same molecule, for all four modifications: inosine, m6A, pseU, and m5C.

I12: Mitochondria contain their own genome, which encodes for a small subset of genes required for mitochondrial function. This project involves examining mitochondrial reads from Nanopore direct-RNA sequencing samples to determine whether they contain any evidence of epi-transcriptomic modification. Is it possible to detect alternative modifications beyond m6A, m5C, pseU, or inosine by examining low-confidence regions within the modification calling model?

I13: How can we detect rare integration events using Nanopore DNA sequencing on a whole-genome wide scale? Employing recently generated data from murine bone-marrow derived macrophages, this project involves designing a comparative rare integration event detector by examining samples with either PBS-treatment, extracellular vesicle-DNA treatment, or lentiviral treatment. The goal is to develop a confident metric of the number of unique integration events within the extracellular-vesicle treated or lentiviral treated samples, in comparison with the PBS-treated samples.

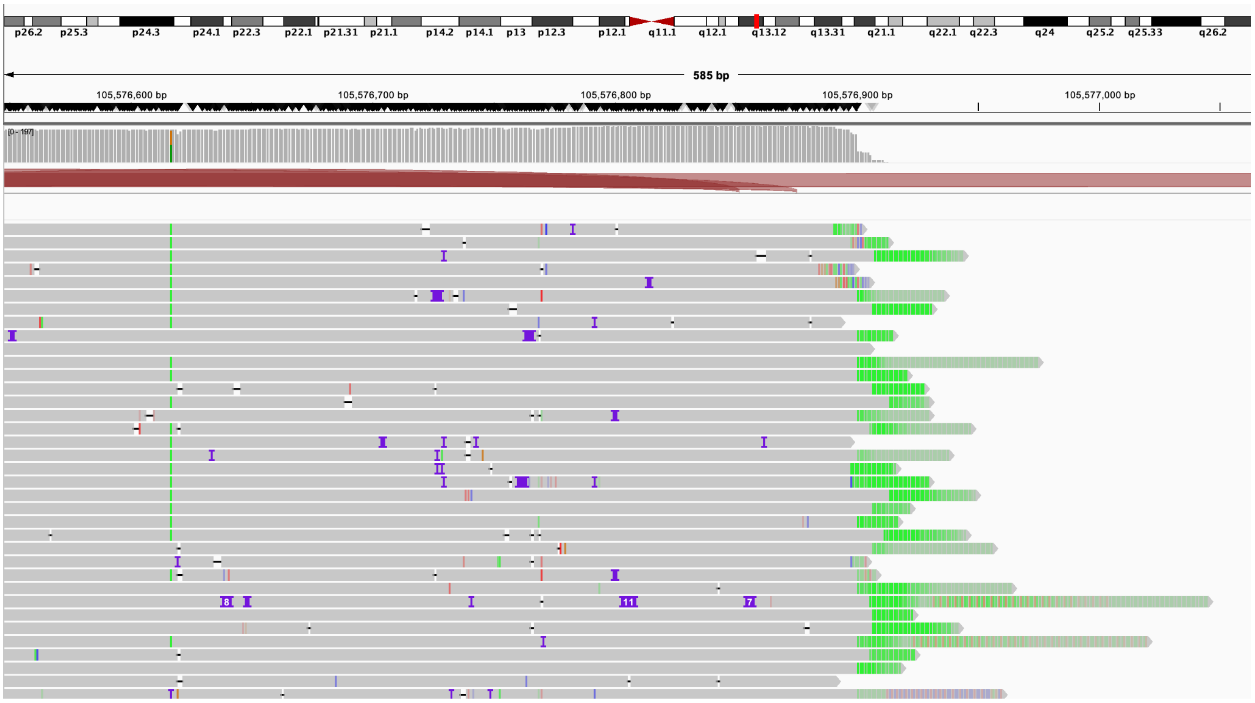

Figure for **I14**; Poly-A tails are in green

I14: Nanopore direct-RNA sequencing captures polyA tails. This information can identify 3’ ends and potential alternative polyadenylation events between samples. This project involves building a tool to identify sites and quantify 3’ alternative polyadenylation in Nanopore direct-RNA sequencing data.

I15: The relationship between genetic variance in a population and the epitranscriptome is poorly understood. This project involves developing a framework for call and perform joint m6A site - single-nucleotide polymorphism analysis to identify associations within high-throughput Nanopore direct-RNA sequencing data. *Please note that this project has already been assigned to a student.*

I16: For a high-coverage gene-locus, identify the number of unique epitranscriptomic molecule states and the proportion of molecules between these states. Each state could be defined as a combination of particular measured modified molecules and canonical modifications. Are there certain states that are more prevalent than others? What are the qualities of these states?

I17: Wastewater sequencing and surveillance are important topics. This project involves analyzing recently generated class samples of wastewater to identify sample composition and eventually perform de novo assembly. This project will aim to incorporate epigenetic information into metagenomic analysis.

I18: Your own idea! Feel free to email thn4005@med.cornell.edu, jip2007@med.cornell.edu, and chm2042@med.cornell.edu.

Logistics: These are individual projects; posters are due according to the syllabus deadline. Please feel free to collaborate with others.

Milestones:
May 2nd, 2025: Project Sheet Release
May 6th, 2025: Office Hours 2:30 - 3:30 p.m. ET
May 8th, 2025: Project Selection Deadline [EOD]
May 13th, 2025: Initial Programming Notebooks + Data Cleaning Comments Due Before Class [with feedback sent post-class]
May 20th, 2025: Open OH + Project Work Session [comments on biological results / progress report due after class by EOD]
May 27th, 2025: Programming Notebooks Collected [for feedback]
June 3rd, 2025: Final Presentation First Draft Due [before class] + Open OH + Project Work Session // Feedback Session
June 7th, 2025: Final Presentation Due
June 10th, 2025: Final Presentations